

The background features a light gray grid. Scattered across the grid are several 3D stick figures in various colors (blue, green, pink, cyan, purple) and poses. On the right side, there is a white sphere with three arrows pointing outwards in green, blue, and red directions.

Progress Update – Capstone Project 2025/2026

Human Pose Estimation

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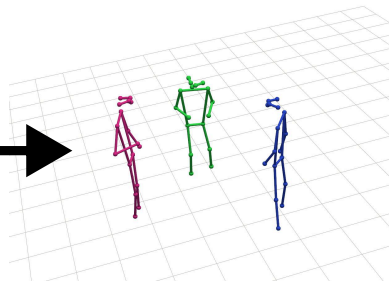
Task – Human Pose Estimation

- ▶ Given:

- Multiple video sequences containing multiple people.
- Recorded using multiple calibrated cameras.

- ▶ Reconstruct:

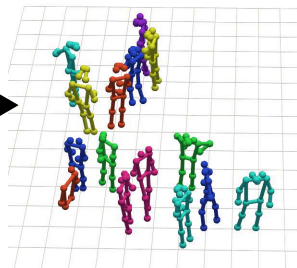
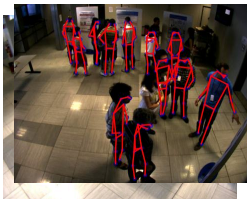
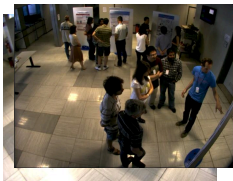
- 3D coordinates of skeletal joint for each subject.
- Movement of skeletal joint over time.



How am I trying to solve it?

- ▶ Summary:

- Perform single-view single-frame 2D estimation using pretrained model.
- Reconstruct 3D geometry from multiple camera views via triangulation.
- Apply Kalman filtering to smooth out noise.



What makes this difficult?

- Challenges:
 - Occlusions can make some joints unobserved.
 - Matching predictions between multiple views.
 - Matching predictions across time.
 - Noisy 2D estimates make the above more difficult.
 - 2D pose estimator often misses some people.



Current Approach – Extended Kalman Filter

► Summary:

→ Keep internal 3D state of all currently tracked persons.

Repeat for each frame:

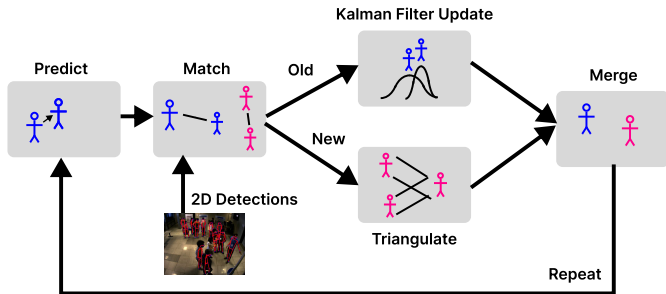
→ Predict next frame state using current state.

→ Match predictions with detections.

→ Update internal 3D state using detections.

→ Associate unmatched detections between views.

→ Create new internal state using triangulation.



What to do next?

- ▶ Learn filter parameters from data.
 - Currently, just guessed manually.
- ▶ Add constraint that forces consistency of joint lengths.
 - To improve results if person is partially occluded.
- ▶ Try using a particle filter with occlusion aware likelihood.
 - More expressive than Kalman filter.
 - Could accurately model occlusions.

